

1. (MC1) $y = 2x - 3 \iff 2x - y = 3$. **Answer:** $2x - y = 3$.
2. (MC2) Same slope 2; different intercepts unless $k = 1$. **Answer:** No solution if $k \neq 1$, infinitely many if $k = 1$.
3. (MC3) Point-slope uses given point and slope directly. **Answer:** $y + 5 = 3(x - 2)$.
4. (MC4) For $(2, 4)$: Above $-\frac{1}{2}(2) + 4 = 3$ (since $4 > 3$) and on/below $2(2) + 1 = 5$ (since $4 \leq 5$). Others fail one of the two. **Answer:** $(2, 4)$.
5. (MC5) $x \in [-3, 3]$. Endpoints map to $y = 5$ and $y = -1$, so range $[-1, 5]$. **Answer:** Domain $[-3, 3]$, Range $[-1, 5]$.
6. (MC6) A point on a graph satisfies its equation; the intersection lies on both, so it satisfies both simultaneously.
7. (MC7) $(0, 0)$ satisfies $0 \leq \frac{1}{2} \cdot 0 + 1$. It fails $y \geq -x + 6$. **Answer:** $y \leq \frac{1}{2}x + 1$.
8. **Long 1.**

Solution: (a) Exactly one solution when $m \neq -2$ (different slopes). If $m = -2$, either same line (infinitely many) or distinct parallel (none).
 (b) If $(1, 3)$ lies on $y = mx + b$, then $3 = m \cdot 1 + b \Rightarrow b = 3 - m$.
 (c) At $x = 0$: inequalities compare $y \geq b$ and $y < 5$. The point $(0, 0)$ is in the overlap iff $0 \geq b$ and $0 < 5$; i.e., iff $b \leq 0$ (slope m is irrelevant at $x = 0$).

9. **Long 2.**

Solution: (a) “Decrease 1 in y per +1 in x ” $\Rightarrow$ slope $m = -1$.
 (b) With $m = -1$ and y -intercept 4: $y = -x + 4$. The x -intercept satisfies $0 = -x + 4 \Rightarrow x = 4$. The “average of intercepts” condition is consistent with linearity (midpoint on the segment of intercepts also lies on the line).
 (c) Point-slope using $(4, 0)$: $y - 0 = -1(x - 4)$. Advantage: pins the line with a simple point and known slope, minimizing algebra in checks.

10. **Long 3.**

Solution: (a) The second equation is $4x - 2y = 10 \iff y = 2x - 5$. Together with $y = 3x - 5$, the slopes differ (3 vs 2) so there is exactly one intersection; the student's $(x, y) = (-10, -35)$ does *not* satisfy $y = 2x - 5$ (since $-35 \neq -25$).
 (b) Error: $-2(3x - 5) = -6x + 10$, not $-6x - 10$. Correct: $4x - 6x + 10 = 10 \Rightarrow -2x = 0 \Rightarrow x = 0$, $y = -5$.
 (c) **Exactly one** solution $(0, -5)$ (non-parallel lines intersect once).